

A Vision Based Framework for Stress Detection in Programming Using Eyebrow Features

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Abstract—Programming tasks can be mentally demanding for students, often causing stress that may affect their performance and learning outcomes. In this study, we developed a vision-based system that detects stress by analyzing the eyebrow region from classroom video recordings. The system works in two stages. First, it determines whether a student is stressed or not stressed. If stress is detected, the second stage classifies the level of stress as low, moderate, or high. To evaluate the system, we tested both traditional machine learning methods and deep learning models, including MobileNetV2 and CNN. Unlike many existing approaches that analyze the entire face or combine multiple data sources, our method focuses only on eyebrow features. This reduces computational complexity while still capturing enough information to detect stress accurately. The experimental results were very encouraging. Deep learning models, especially MobileNetV2 and CNN, performed consistently well, achieving up to 92% accuracy in both stages. These findings show that eyebrow-based analysis is an effective and non-intrusive approach for real-time stress monitoring in educational environments.

Index Terms—Stress recognition, Facial feature extraction, Eyebrow-based analysis, Deep learning, Machine learning, Human-computer interaction

I. INTRODUCTION

Programming is a mentally demanding activity that often causes stress in students [1]. This stress can affect both their academic performance and their ability to learn effectively. Traditional methods for detecting stress, such as self-reports and physiological sensors, have several limitations [2]. Self-reports are subjective and may not accurately capture stress in real time, while sensor-based methods can be intrusive and usually require controlled environments. As a result, these approaches are not well suited for classroom use, creating a need for a more practical and non-intrusive solution.

In this work, we propose a computer vision-based approach for detecting stress using eyebrow features. The system is

designed to identify stress efficiently and operates in two stages. In the first stage, it determines whether a student is stressed or not stressed. In the second stage, it classifies the level of stress into low, moderate, and high categories. This structured approach separates stress detection from stress level estimation, allowing for more precise analysis. To improve performance, the framework incorporates both traditional machine learning models and deep learning architectures.

Programming activities such as debugging code, understanding logic, and working under time constraints can significantly increase mental workload and cognitive strain [3]. This can negatively affect learning outcomes and reduce students' ability to perform effectively. In addition to academic performance, stress can also impact students' emotional well-being and their long-term interest in programming. Continuous exposure to high cognitive load may lead to frustration, lower confidence, and reduced engagement in learning activities. Therefore, early stress detection is essential for enabling timely interventions and creating more adaptive and supportive learning environments.

With the growing use of technology in education, there is an increasing need for automated and non-intrusive stress monitoring systems. Computer vision-based approaches provide an effective solution because they are scalable and easy to deploy in real-world educational settings. Recent studies have shown that specific facial regions, particularly the eyes and eyebrows, are strong indicators of stress and cognitive load [4]. Subtle changes in eyebrow movement and eye tension can reveal mental effort and emotional states. By focusing only on these localized regions, the system reduces computational complexity while still capturing meaningful stress-related information.

Recent advances in deep learning and edge computing have made it possible to deploy real-time stress detection systems in

practical environments. These systems can operate efficiently with limited computational resources while maintaining high accuracy. By combining feature extraction techniques with optimized machine learning and deep learning models, stress monitoring can be integrated seamlessly into modern educational environments without disrupting the learning process.

II. RELATED WORK

Programming tasks place a high cognitive demand on students and often lead to increased stress, which can negatively affect both learning and performance. Previous studies have examined stress during programming using techniques such as eye tracking and behavioral analysis to understand how students respond to challenging tasks. Research shows that less experienced programmers are more likely to experience stress when dealing with complex code and debugging activities [5]. Traditional stress detection methods, including self-reports and physiological sensors, have several limitations. Self-reports are subjective and may not accurately capture stress in real time, while sensor-based approaches are intrusive and often require controlled environments. These drawbacks limit their practical use in classroom settings [6]. Recent advances in computer vision and affective computing have enabled non-intrusive stress detection using facial expressions [7]. Most existing systems analyze the entire face or combine multiple modalities, which can increase computational cost and reduce suitability for real-time applications. Studies have shown that the eye and eyebrow regions contain valuable information related to stress and cognitive load [8], making them effective indicators of emotional and mental states. Machine learning techniques such as Support Vector Machines, Random Forest, and K-Nearest Neighbors have been widely used for stress classification [9]. More recently, deep learning models including Convolutional Neural Networks (CNN) and MobileNetV2 have demonstrated improved performance by automatically learning discriminative features from visual data [10].

Building on this work, the proposed system focuses specifically on eyebrow-region features and employs a two-stage classification framework. The first stage identifies whether a student is stressed, and the second stage classifies the stress level as low, moderate, or high. By combining efficient feature extraction with machine learning and deep learning models, the system provides an accurate, real-time, and non-intrusive solution for stress monitoring in educational environments.

III. PROPOSED METHODOLOGY

A. Dataset Collection and Preparation

The dataset used in this study was created to analyze stress levels in students during programming activities. It consists of video recordings from 38 undergraduate students who were still in the learning stage and had not yet developed expert-level programming skills [11]. The recordings were captured in a classroom environment while the students worked on programming tasks. Each participant was recorded for approximately 2 to 3 minutes, providing facial video data that reflected their expressions and reactions while solving coding problems.

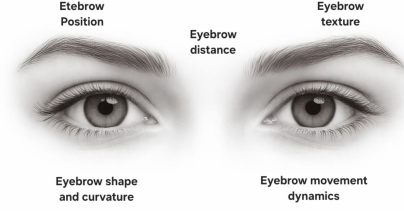


Fig. 1. Grayscale eye image showing visual features used for stress detection.

The recorded videos were first converted into individual frames so that a large number of facial images could be obtained. Each frame was then analyzed using a detection method to identify the eyebrow region, which is the main area used in this study to detect stress. Images that were blurred, unclear, or of poor quality were removed to maintain the quality of the dataset. The remaining images were resized and standardized so that all samples had the same format. This preprocessing step helps the models perform more consistently and improves overall stability. Both the facial images and the extracted eyebrow regions play an important role in analyzing stress-related changes in facial expressions.

Dataset Component	Count
Video Recordings	38
Facial Image Frames	4,278
Eyebrow-region Images	4,278

TABLE I
SUMMARY OF FINAL DATASET

B. Preprocessing and Feature Extraction

For each video frame, the face is first detected, and key facial landmarks are identified to accurately locate the eyebrow region [12]. The eyebrow images are then extracted, resized to a uniform size, and normalized so that all samples have a consistent appearance. This preprocessing step helps both traditional machine learning and deep learning models analyze the images more effectively and improves their ability to detect stress-related patterns.

C. Two-Stage Classification Framework

The proposed system follows a two-stage classification approach for effective stress detection and stress level analysis.

1) *Stage 1: Stress Detection:* In the first stage, the system analyzes eyebrow images and classifies them into two categories: stressed and non-stressed. To perform this task, both traditional machine learning methods and deep learning models are used. The models evaluated in this stage include Support Vector Machine (SVM), Random Forest, Gradient Boosting Classifier, Extreme Gradient Boosting (XGBoost), Convolutional Neural Networks (CNN), and MobileNetV2. These models are trained to determine whether the eyebrow patterns indicate stress or not.

2) *Stage 2: Stress Level Classification*: In the second stage, the samples identified as stressed in Stage 1 are analyzed further and classified into three stress levels: low stress, moderate stress, and high stress.

This two-stage approach improves the overall classification process by separating stress detection from stress intensity estimation. By handling these as two distinct steps, the system is able to classify stress levels more accurately and reliably.

D. System Workflow

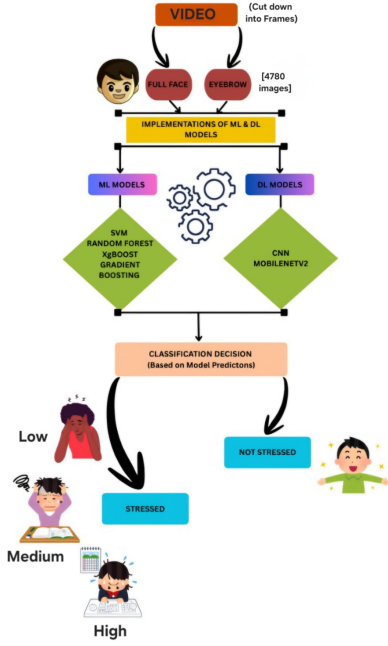


Fig. 2. Architecture of the Proposed Stress Detection and Stress Level Analysis Framework

The overall process begins by recording students while they solve programming problems. The videos are divided into individual frames, and the eyebrow region is extracted from each frame. After preprocessing, these images are passed through the classification models to detect whether stress is present and to estimate its intensity.

The framework operates in two consecutive steps. First, it determines whether the student is experiencing stress. If stress is detected, the same sample is analyzed further and assigned to one of three categories: low, moderate, or high stress. The final output indicates both the presence of stress and its intensity, providing a clearer understanding of how programming tasks affect students during the learning process.

E. Model Evaluation

We measured the effectiveness of the models using standard classification metrics such as accuracy, precision, recall, F1-score, and confusion matrices. These metrics provide a clear picture of how well each model identifies stress and distinguishes between different stress levels. These metrics

provide a comprehensive assessment of how well the models perform in both stages of classification. Accuracy measures the overall correctness of the predictions, while precision and recall indicate how effectively the model identifies the correct stress categories. The F1-score provides a balanced measure of precision and recall, and the confusion matrix offers a detailed view of the classification results. Each stage of the two-stage framework is evaluated using these metrics to provide a clear understanding of the model's performance.

IV. RESULTS AND ANALYSIS

To understand how well the system works, we tested two different approaches: one that focuses only on the eyebrow region and another that analyzes the full face. This comparison helps determine whether a smaller facial region can provide reliable stress-related information [13]. Similar to other vision-based stress detection systems, the proposed framework follows a two-stage classification process. In the first stage, the system determines whether the individual is stressed or not stressed. In the second stage, machine learning and deep learning models are used to classify the level of stress as low, moderate, or high. By comparing eyebrow-based analysis with full-face analysis, the study assesses how well the proposed method performs relative to existing approaches. The results show that the proposed framework is effective in accurately classifying stress levels while maintaining a simpler and more efficient design.

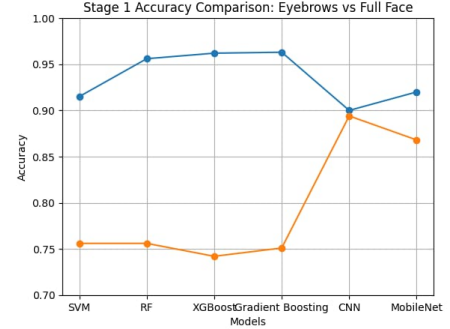


Fig. 3. Stage 1 Accuracy Comparison: Eyebrows vs Full Face

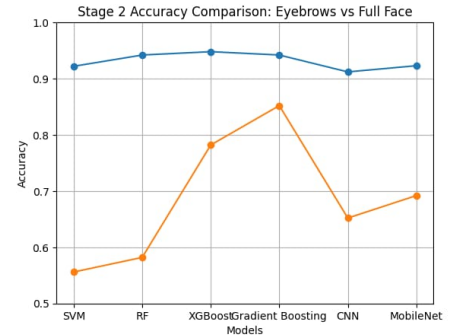


Fig. 4. Stage 2 Accuracy Comparison: Eyebrows vs Full Face

Among the deep learning models tested, MobileNetV2 consistently produced the strongest results. Its transfer learning capability allows it to capture subtle visual changes associated with stress while requiring relatively little computational power, making it suitable for real-time applications. In addition, recent research by Khomidov (2024) demonstrated that lightweight deep learning models can detect stress in real time while requiring relatively low computational resources. The results of this study further confirm that efficient transfer learning models like MobileNetV2 are well suited for real-time stress detection and stress level classification.

A. Eyebrow-Based Stress Recognition Results

TABLE II
STAGE 1 MODEL PERFORMANCE COMPARISON (STRESSED VS NOT STRESSED)

Model	Accuracy	Precision	Recall	F1-Score
SVM	0.915	0.93	0.91	0.92
Random Forest	0.956	0.96	0.96	0.96
XGBoost	0.961	0.96	0.96	0.96
Gradient Boosting	0.961	0.96	0.96	0.96

TABLE III
STAGE 2 MODEL PERFORMANCE COMPARISON (LOW, MEDIUM, HIGH STRESS)

Model	Accuracy	Precision	Recall	F1-Score
SVM	0.920	0.92	0.92	0.92
Random Forest	0.941	0.94	0.94	0.94
XGBoost	0.947	0.95	0.95	0.95
Gradient Boosting	0.940	0.94	0.94	0.94

TABLE IV
STAGE 1 DEEP LEARNING MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score
CNN	0.90	0.91	0.90	0.90
MobileNetV2	0.92	0.92	0.92	0.91

TABLE V
STAGE 2 DEEP LEARNING MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score
CNN	0.91	0.91	0.90	0.90
MobileNetV2	0.92	0.92	0.92	0.92

In the eyebrow-focused approach, only the region above the eyes is used for analysis. Even though this area contains limited visual information, the models were able to identify stress patterns with high accuracy, showing that eyebrow movements alone provide meaningful cues about a student's mental state. Traditional machine learning models such as Support Vector Machine (SVM), Random Forest, Gradient Boosting, and XGBoost were trained using manually extracted features. In contrast, deep learning models such as CNN and MobileNetV2 learned stress-related features directly from the eyebrow images. Previous studies, including Ramesh (2021), have shown that the eyebrow and eye regions contain useful information

for detecting stress [14]. In the first stage, the models were able to distinguish effectively between stressed and non-stressed students. Traditional models such as Random Forest produced strong classification results, while deep learning models automatically learned important eyebrow patterns associated with stress. MobileNetV2 achieved particularly strong performance while requiring relatively low computational resources. In the second stage, the stressed samples were further classified into three categories: low stress, moderate stress, and high stress. The results indicate that eyebrow features contain sufficient information to estimate stress intensity accurately. Consistent with findings reported by Zhang (2020), deep learning models performed better than traditional machine learning methods in multi-class stress classification [15].

B. Full-Face Stress Recognition Results

TABLE VI
STAGE 1 ULTRA BASELINE MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score
SVM	0.756	0.57	0.76	0.65
Random Forest	0.756	0.57	0.76	0.65
XGBoost	0.742	0.70	0.74	0.71
Gradient Boosting	0.751	0.73	0.75	0.74

TABLE VII
STAGE 2 ULTRA BASELINE MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score
SVM	0.555	0.34	0.55	0.42
Random Forest	0.581	0.36	0.58	0.45
XGBoost	0.781	0.79	0.78	0.77
Gradient Boosting	0.852	0.85	0.85	0.85

TABLE VIII
STAGE 1 PERFORMANCE COMPARISON ON FULL-FACE IMAGES

Model	Accuracy (%)	Precision	Recall	F1-score
CNN	89.27	0.85	0.86	0.86
MobileNetV2	86.83	0.93	0.73	0.78

TABLE IX
STAGE 2 PERFORMANCE COMPARISON ON FULL-FACE IMAGES

Model	Accuracy (%)	Precision	Recall	F1-score
CNN	65.16	0.49	0.64	0.54
MobileNetV2	69.03	0.73	0.72	0.70

The full-face method considers additional facial features such as the mouth, forehead, and jawline. These regions can reveal expressions linked to stress, but they also increase the amount of data the model must process [16]. In the first stage, using the complete facial image helped the models distinguish more effectively between stressed and non-stressed students because additional facial features provided more information for classification. This allowed both machine learning and

deep learning models to learn stress-related patterns more accurately. In the second stage, full-face analysis improved the estimation of stress intensity by capturing additional indicators such as jaw tension, pressed lips, and changes in the forehead. These results suggest that analyzing the entire face is an effective approach for classifying stress levels [17], consistent with findings reported by Khomidov (2024), and can contribute to the development of more accurate stress detection systems.

C. Comparative Analysis

TABLE X

COMPARATIVE ANALYSIS OF EYEBROW-BASED AND FULL-FACE STRESS RECOGNITION (STAGE 1)

Approach	Model	Accuracy	Precision	F1-Score
Eyebrow-Based	MobileNetV2	0.92	0.92	0.91
Full-Face	MobileNetV2	0.86	0.93	0.78

TABLE XI

COMPARATIVE ANALYSIS OF EYEBROW-BASED AND FULL-FACE STRESS RECOGNITION (STAGE 2)

Approach	Model	Accuracy	Precision	F1-Score
Eyebrow-Based	MobileNetV2	0.92	0.92	0.92
Full-Face	MobileNetV2	0.69	0.73	0.70

The comparison shows that focusing only on the eyebrow region offers a good balance between accuracy and efficiency. Despite using a smaller portion of the face, the eyebrow-based model performed as well as, and in some cases better than, the full-face approach. Because it focuses only on the eyebrow region, the method requires fewer resources and is well suited for real-time use in educational environments. Although full-face analysis provides additional information about facial expressions, eyebrow-based analysis alone is capable of detecting stress with high accuracy while maintaining a simpler and faster framework. Among the models evaluated, MobileNetV2 delivered particularly strong performance, achieving high accuracy while using relatively low computational power. The study also demonstrates that the two-stage classification approach is effective. In the first stage, the system determines whether a student is stressed or not stressed. In the second stage, the stressed samples are further classified into low, moderate, and high stress levels. Overall, the findings confirm that eyebrow features contain sufficient information for reliable stress detection and stress level estimation. At the same time, full-face analysis can provide additional cues that may further improve performance. These results highlight the potential of vision-based stress monitoring systems for real-time and non-intrusive use in educational settings.

While the results are encouraging, the study has several limitations. The dataset was collected from a single institution, which may reduce generalizability. In addition, variations in lighting, head movement, and camera positioning can affect performance in real-world settings. In addition, stress labeling is not always straightforward, which may introduce inconsistencies in the classification process. Another limitation is that

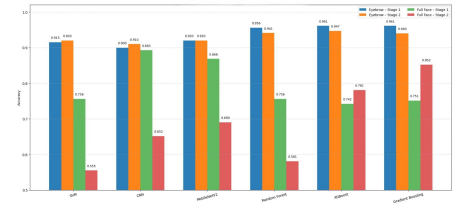


Fig. 5. Final Conclusion Graph, Two-Stage Stress Detection: Eyebrow-Based vs Full-Face Models

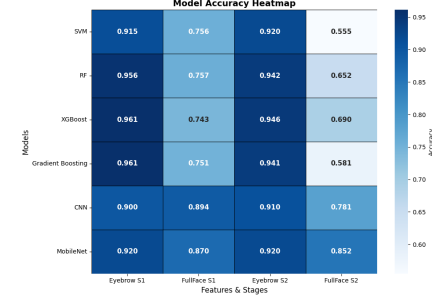


Fig. 6. Comparative heatmap of model accuracy across Stage 1 and Stage 2 using eyebrow and full-face features.

deep learning models often require significant computational resources. Moreover, the current approach analyzes individual frames and does not fully capture temporal changes across video sequences, which could influence performance in real-world applications.

V. CONCLUSION

This study introduced a two-step computer vision framework for recognizing stress in students during programming activities. By analyzing eyebrow features, the system was able to detect both the presence of stress and its severity with strong accuracy [18]. The experimental results show that the approach is highly effective, with some models achieving up to 92% accuracy. Among the models tested, MobileNetV2 delivered particularly strong performance. By concentrating only on the eyebrow region, the system reduces computational cost while still capturing meaningful stress-related patterns.

The two-stage design further improves performance by separating stress detection from stress level classification. In the first stage, the system determines whether a student is stressed or not stressed. In the second stage, stressed samples are classified into low, moderate, and high stress levels. The findings suggest that a lightweight and non-intrusive vision-based system can be used to monitor student stress in educational environments. With more diverse data and temporal modeling, the approach can be made even more robust for practical use [19].

Future work will focus on collecting more diverse datasets and exploring sequence-based models such as LSTM to capture temporal patterns. Additional visual cues may also be incorporated to make the system more robust and effective in real-world applications.

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